

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 03, 2026

VIDEO INFORMATION

Video ID:	12b56586-2a80-44e3-b0dd-db1f0b500fe0
Total Frames:	6113
Frame Rate:	25.0 fps (assumed)
Duration:	244.5 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	1660	✓
RT-DETR Detections	3127	✓
MedSAM2 Detections	3320	✓
Consensus (3-Model Agreement)	6	✓
Risk Classification	Count	Percentage
HIGH RISK	4	66.7%
MEDIUM RISK	2	33.3%
LOW RISK	0	0.0%
TOTAL ANALYZED	6	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 272 frames)



Feature	Value	Interpretation
Frame Index	3700	Frame 3700 of 6113
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	MALIGNANT_P OLYP	Type confidence: 92.0%
Redness Score	0.191	Color-based vascularization
Relative Radius	6.8%	Polyp size as % of frame diagonal
Texture Score	0.521	Surface morphology
Vessel Visibility	0.561	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	MALIGNANT_P OLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	82.0%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #2 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 53 frames)

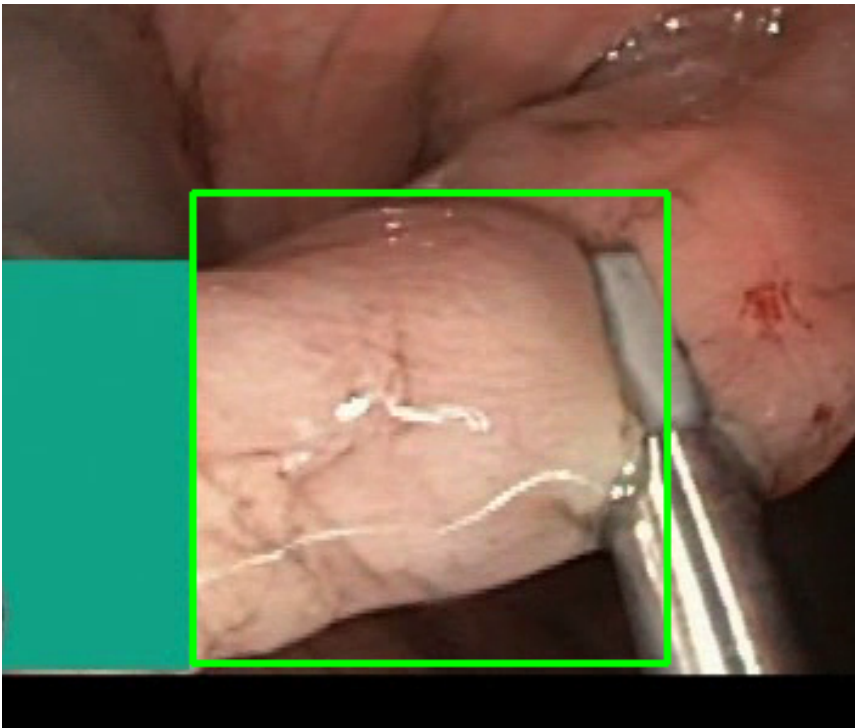


Feature	Value	Interpretation
Frame Index	1983	Frame 1983 of 6113
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	MALIGNANT_POLYP	Type confidence: 92.0%
Redness Score	0.197	Color-based vascularization
Relative Radius	6.3%	Polyp size as % of frame diagonal
Texture Score	0.716	Surface morphology
Vessel Visibility	0.399	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	85.4%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #3 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 386 frames)



Feature	Value	Interpretation
Frame Index	2550	Frame 2550 of 6113
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 92.0%
Redness Score	0.169	Color-based vascularization
Relative Radius	8.1%	Polyp size as % of frame diagonal
Texture Score	0.614	Surface morphology
Vessel Visibility	0.385	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	83.5%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #4 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 48 frames)



Feature	Value	Interpretation
Frame Index	3407	Frame 3407 of 6113
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	SERRATED_POLYP	Type confidence: 84.8%
Redness Score	0.126	Color-based vascularization
Relative Radius	37.2%	Polyp size as % of frame diagonal
Texture Score	0.145	Surface morphology
Vessel Visibility	0.290	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	86.2%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #5 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 70 frames)



Feature	Value	Interpretation
Frame Index	3793	Frame 3793 of 6113
Detection Method	Detector Consensus (MedSAM2 / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	MALIGNANT_POLYP	Type confidence: 92.0%
Redness Score	0.217	Color-based vascularization
Relative Radius	6.6%	Polyp size as % of frame diagonal
Texture Score	0.412	Surface morphology
Vessel Visibility	0.525	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	MALIGNANT_POLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	81.8%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #6 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 132 frames)



Feature	Value	Interpretation
Frame Index	3445	Frame 3445 of 6113
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	MALIGNANT_POLYP	Type confidence: 92.0%
Redness Score	0.137	Color-based vascularization
Relative Radius	26.3%	Polyp size as % of frame diagonal
Texture Score	0.175	Surface morphology
Vessel Visibility	0.478	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	81.2%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:

A total of 6 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 4 (66.7%)
- MEDIUM RISK polyps: 2 (33.3%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.